SIMATS ENGINEERING
ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1506

COURSE OUTCOME COVERED
CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 30**

Question Count: 10

Code of Conduct

I **Dinesh V (192524309)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Dinesh V**

Assessment Tasks
Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.

Answer:

A unified big data architecture should integrate customer information from online stores, mobile applications, physical stores, CRM systems, loyalty programs, and social-media interactions.

Proposed architecture:

• Data sources: websites, mobile apps, POS systems, CRM, loyalty cards and social-media platforms.
• Ingestion layer: Apache Kafka or cloud streaming services for real-time events, with batch ingestion for historical data.
• Storage layer: a scalable data lake using cloud object storage such as Amazon S3, Azure Data Lake Storage or Google Cloud Storage. A data warehouse can store cleaned, structured data for reporting.
• Processing layer: Apache Spark/Flink for distributed batch and stream processing.
• Unified customer profile: combine customer IDs, purchases, browsing history, preferences and interaction data using data integration and identity-resolution techniques.
• Analytics/ML layer: customer segmentation, churn prediction, purchase prediction and recommendation models.
• Serving layer: APIs and low-latency databases deliver recommendations to websites, mobile apps and point-of-sale systems.
• Governance and security: encryption, access control, data quality checks, auditing and privacy policies.

Real-time customer events can be processed immediately to update customer profiles and generate personalized recommendations. For example, if a customer repeatedly views laptops online, the system can recommend related accessories or suitable laptop models. This architecture reduces data silos and supports consistent, real-time decision-making across channels.

2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.

Answer:

Distributed computing divides a large scientific workload among multiple computers or processing nodes. Models such as Hadoop MapReduce, Apache Spark and distributed cloud computing are effective for genome analysis, climate modelling, astronomy, physics simulations and large-scale experiments.

In a centralized system, one machine performs most computations, so CPU, memory and storage become bottlenecks. In a distributed system, a large dataset is partitioned across nodes and computations are executed in parallel.

Parallel processing improves efficiency because independent tasks can run simultaneously. Spark can also keep frequently used data in memory, reducing repeated disk access. Distributed systems provide horizontal scalability: additional nodes can be added when data or workload increases.

For example, in genome analysis, millions of DNA sequences can be divided among several nodes, allowing different sequences to be processed simultaneously. Fault tolerance mechanisms can recover failed tasks or replicate data.

Therefore, distributed computing provides higher throughput, better scalability and improved resource utilization than a single centralized system, although it introduces challenges such as network communication, synchronization, data partitioning and distributed-system management.

3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.

Answer:

A fraud detection framework should combine real-time transaction streams, historical transaction data and machine-learning models.

Framework:

1. Data sources: payment transactions, device information, IP addresses, merchant details, location, transaction history and user behaviour.
2. Streaming layer: Apache Kafka can capture transactions continuously.
3. Stream processing: Apache Flink or Spark Streaming can calculate real-time features such as transaction frequency, unusual location and spending amount.
4. Storage: a data lake stores historical transactions, while a database stores recent customer and risk information.
5. Predictive analytics: supervised models such as Logistic Regression, Decision Trees, Random Forest or Gradient Boosting can classify transactions as legitimate or suspicious. Anomaly-detection models can identify previously unseen fraud patterns.
6. Risk engine: assign a fraud-risk score to every transaction.
7. Response: allow low-risk transactions, request additional authentication for medium-risk transactions, and block or investigate high-risk transactions.
8. Monitoring: continuously evaluate precision, recall, false positives and false negatives.

Streaming processing enables a transaction to be evaluated within seconds instead of waiting for batch processing. Predictive models learn patterns from historical fraud and legitimate transactions. For example, a sudden high-value transaction from a new device and unusual location can receive a high-risk score. Combining real-time features with predictive analytics improves early detection and reduces financial losses.

4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.

Answer:

Multimedia data is challenging because it is large, heterogeneous, unstructured and often generated continuously. High-resolution video requires significant storage and bandwidth, while images and audio require specialized processing. Metadata, different file formats, duplication, privacy and fast retrieval are additional challenges.

Suitable architecture:

- Object storage/data lake: Amazon S3, Azure Blob Storage, Google Cloud Storage or HDFS for large multimedia files.
- Distributed processing: Apache Spark can process metadata and large collections of files in parallel.
- Streaming: Kafka can ingest continuously generated video/audio events.
- Databases: NoSQL databases such as MongoDB or Cassandra can store flexible metadata and indexes.
- Compression: codecs such as H.264/H.265 for video and appropriate audio/image compression reduce storage requirements.
- Content delivery: CDN services can provide faster delivery of frequently accessed media.
- Metadata cataloguing: store file type, timestamp, location, owner and content-related metadata for efficient search.
- Security: encryption, authentication, authorization and access logging protect sensitive media.

For example, a video-streaming platform can store original videos in object storage, generate compressed versions for different resolutions, store metadata in a database and use a CDN for delivery. This approach improves scalability, availability and retrieval performance.

5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.

Answer:

A disaster recovery strategy should ensure that data and services remain available during hardware failure, software errors, cyberattacks or regional cloud outages.

Strategy:

- Identify critical services and define Recovery Point Objective (RPO) and Recovery Time Objective (RTO).
- Use multiple availability zones or regions.
- Replicate critical datasets across independent storage locations.

- Maintain automated and versioned backups.
- Use redundant compute nodes and load balancing.
- Deploy applications using containers or orchestration platforms such as Kubernetes.
- Use health monitoring to detect failures.
- Configure automated failover to a healthy node, zone or region.
- Maintain documented recovery procedures and conduct regular disaster-recovery tests.
- Protect backups using encryption and access control.

Replication ensures that a failure of one storage location does not destroy the only copy of data. Redundant compute resources keep services running when a node fails. Automated failover reduces recovery time because traffic can be redirected without waiting for manual intervention.

For example, if a primary cloud region becomes unavailable, the system can redirect users to a replicated secondary region while maintaining access to critical data. Thus, redundancy, replication and automated failover provide high availability and business continuity.

6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.

Answer:

Poor data quality directly affects analytical accuracy and business decisions. Inconsistent values may represent the same entity differently, duplicates can inflate counts, and missing values can bias statistical and machine-learning results.

Impacts include:

- Incorrect reports and dashboards.
- Biased machine-learning predictions.
- Duplicate customer or transaction counts.
- Incorrect business decisions.
- Increased data-cleaning costs.
- Reduced trust in analytics.

Data governance strategies:

1. Establish data ownership and stewardship responsibilities.
2. Define standard formats, naming conventions and validation rules.
3. Use automated data-quality checks for completeness, accuracy, consistency, uniqueness and validity.
4. Apply deduplication and entity-resolution methods.
5. Handle missing values using appropriate imputation, removal or business rules.
6. Maintain metadata and a data catalogue.
7. Track data lineage so the origin and transformations of data are known.
8. Monitor data quality continuously and generate alerts when thresholds are violated.

9. Apply access control and auditing.

For example, if customer records contain different spellings or multiple IDs for the same person, identity resolution can merge them into a unified profile. Strong governance therefore improves the reliability and reproducibility of analytics.

7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.

Answer:

A scalable recommendation system can combine user behaviour, content information and machine-learning models.

Architecture:

- Collect events such as searches, clicks, watch time, likes, skips and ratings.
- Use Kafka or another streaming platform to collect events continuously.
- Store historical events in a cloud data lake.
- Use Spark for distributed feature engineering and model training.
- Build user and item profiles using viewing history, genre preferences and content characteristics.
- Apply collaborative filtering, content-based filtering or hybrid recommendation models.
- Use a low-latency serving database/cache to return recommendations quickly.
- Continuously update recommendations using recent streaming behaviour.
- Monitor metrics such as click-through rate, watch time, completion rate and user retention.

User behaviour analytics identifies what users prefer and how their interests change. Distributed processing allows billions of interactions to be processed efficiently and enables large-scale model training.

For example, if a user watches several science-fiction films, the system can recommend similar movies. If the user's behaviour changes toward documentaries, recent activity can influence the next recommendation list. Personalized recommendations increase content discovery, watch time and user engagement.

8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

Answer:

Cloud computing provides on-demand computing, storage and analytics resources for big data systems. Cloud platforms offer services for data lakes, warehouses, distributed processing, streaming, machine learning and monitoring.

Advantages:

• Scalability: compute and storage can be increased or decreased according to workload.
• Cost management: pay-as-you-go models reduce the need for large upfront infrastructure investments.
• Flexibility: organizations can select different compute, storage, database and analytics services.
• Availability: cloud platforms can provide multiple availability zones and regions.
• Faster deployment: managed services reduce infrastructure-management effort.
• Security tools: cloud providers offer encryption, identity management, logging and security monitoring.

Limitations:
• Costs can become high if resources are poorly monitored or continuously overprovisioned.
• Vendor lock-in can make migration difficult.
• Data transfer and network costs can increase for large datasets.
• Security remains a shared responsibility; incorrect configuration can expose data.
• Compliance and data residency requirements may restrict where data can be stored.
• Dependence on network connectivity and cloud-provider availability can affect operations.

Therefore, cloud computing is highly suitable for big data because it provides elastic resources and managed services, but organizations must use cost controls, strong security, governance and multi-cloud or portability strategies where appropriate.

9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.
Answer:
A real-time transportation analytics system should collect data continuously from vehicles, roads, warehouses and external services.

Framework:
• Data sources: GPS devices, vehicle sensors, traffic APIs, weather services, fuel sensors and delivery records.
• Ingestion: Kafka or cloud streaming services collect real-time events.
• Stream processing: Spark Streaming or Flink processes location, speed, fuel and traffic information.
• Storage: a data lake stores historical records, while fast databases support current vehicle information.
• Analytics: calculate estimated arrival time, congestion, fuel consumption, route cost and delivery performance.
• Optimization: route algorithms select efficient paths based on traffic, distance, delivery priority and vehicle capacity.
• Visualization: dashboards display vehicle locations, alerts, ETAs and route performance.
• Predictive analytics: predict delays, demand and vehicle maintenance requirements.

Real-time traffic information enables the system to reroute vehicles when congestion or accidents occur.

Historical fuel and route data can identify fuel-inefficient patterns. Predictive analytics can estimate delays and help managers make proactive decisions.

For example, if a delivery truck is approaching a congested road, the system can calculate an alternative route and update its ETA. This reduces travel time and fuel consumption while improving delivery reliability.

10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Answer:

Big data creates ethical and legal challenges because organizations may collect large amounts of personal and behavioural information. Major concerns include privacy, consent, surveillance, discrimination, unauthorized access, data breaches and unclear data ownership.

Key challenges:

- Privacy: excessive collection can expose sensitive personal information.
- Consent: users may not fully understand how their data is collected or used.
- Security: large datasets are attractive targets for cyberattacks.
- Bias: biased training data can produce discriminatory decisions.
- Transparency: complex analytics may make automated decisions difficult to explain.
- Data retention: keeping data longer than necessary increases risk.
- Regulatory compliance: organizations must follow applicable privacy and data-protection requirements.

Recommended policies:

1. Collect only necessary data and define a legitimate purpose.
2. Obtain informed consent where required and provide clear privacy notices.
3. Use encryption, access controls, authentication and security monitoring.
4. Apply anonymization or pseudonymization where appropriate.
5. Establish data retention and secure deletion policies.
6. Conduct regular privacy, security and algorithmic-bias assessments.
7. Maintain data lineage, audit logs and accountability.
8. Provide mechanisms for users to understand and exercise applicable data rights.
9. Train employees on responsible data handling.
10. Establish incident-response and breach-notification procedures.

For example, before using customer behaviour to train a recommendation model, an organization should define the purpose, protect personal information and apply appropriate privacy controls. Responsible governance strengthens regulatory compliance and user trust while allowing organizations to gain value from big data.

Class Test Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity